MATRICA

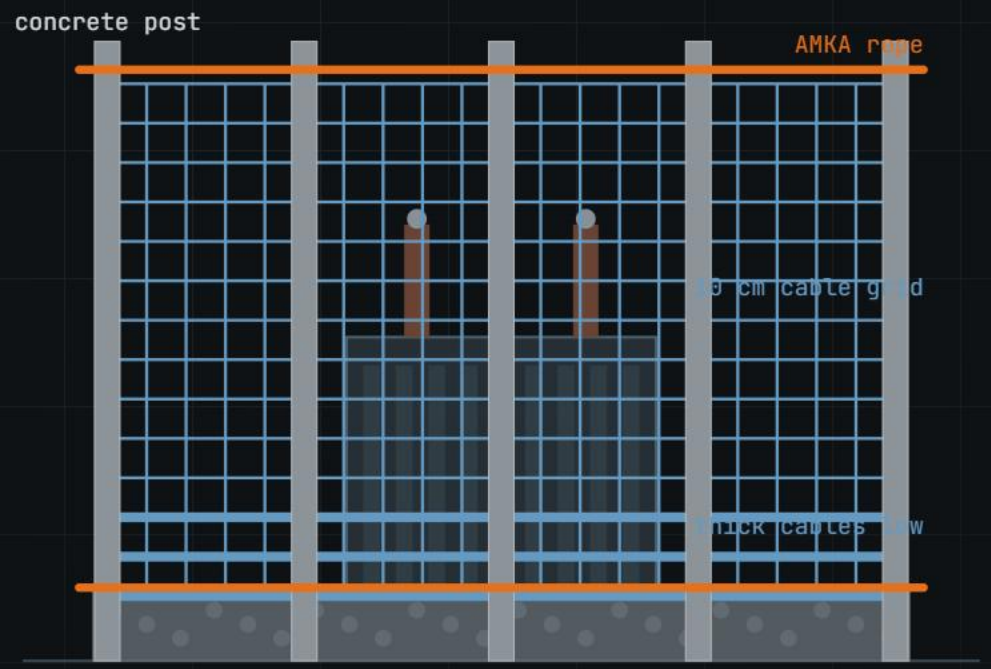
Grid waste becomes grid shield.

We take the cables and concrete poles grid companies throw away, and build shields around small, critical grid equipment.

WE TAKE Cable offcuts + Concrete poles → Shield

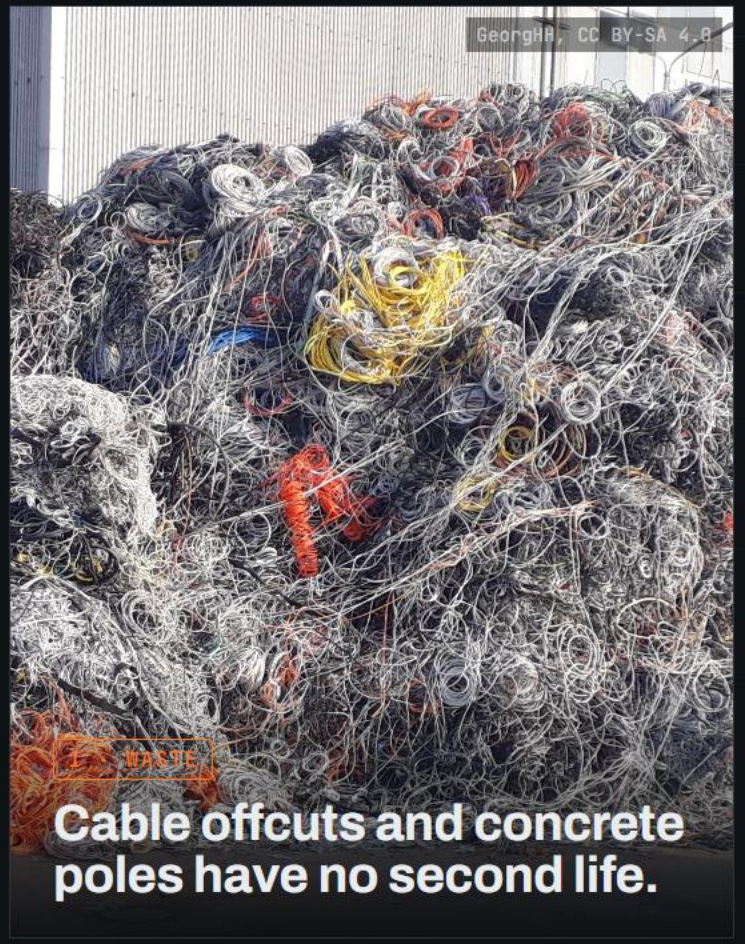
Climathon 2026 · RTU

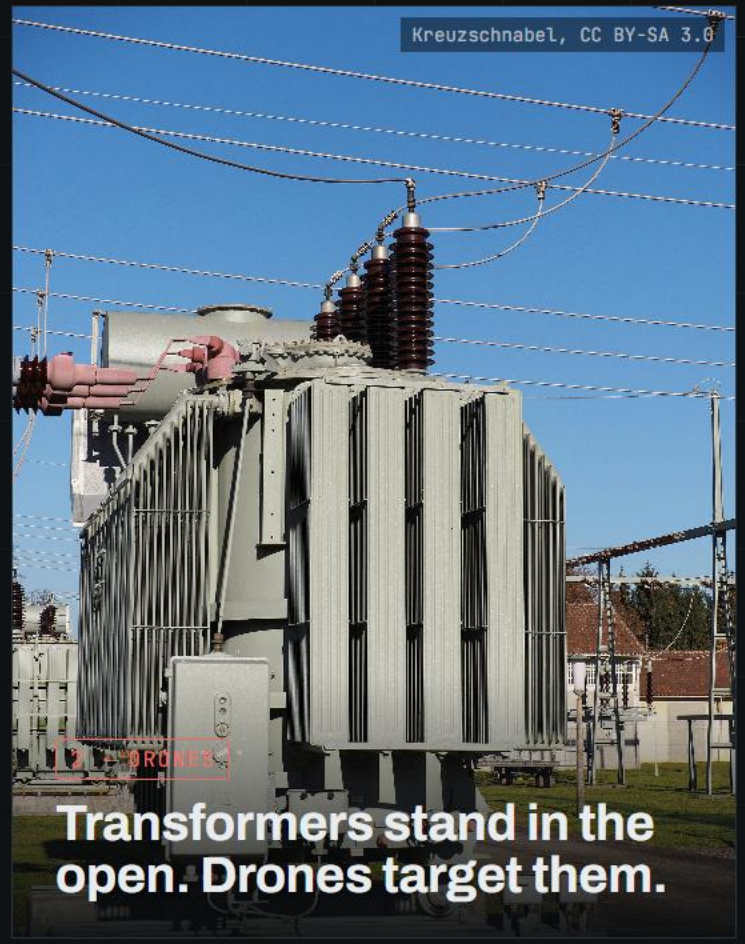
Challenge: AS "Sadales tīkls"



THE PROBLEM

Three problems, one place.







BEFORE

Today: transformers stand unprotected.



MATRICA

Photo: Tiia Monto, CC BY-SA 4.0

AFTER · WITH MATRICA

Reused concrete posts + a net from cables.

Crushed concrete forms the base. Same real substation; the shield is an AI visualisation (Qwen-Image-Edit).



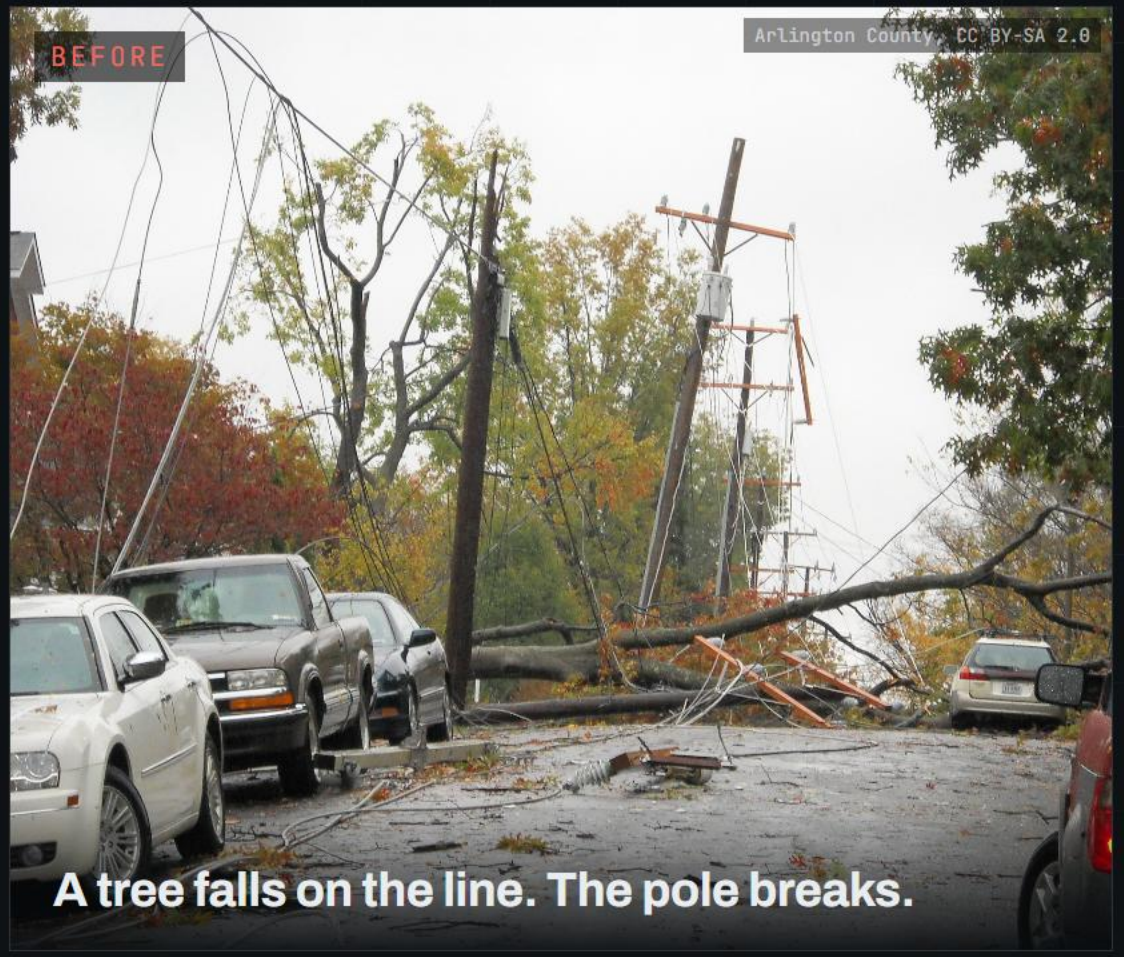
MATRICA

Climathon 2026 · Gaia Tech Challenge

Photo: Tiia Monto, CC BY-SA 4.0 · AI edit: Qwen-Image-Edit

STORMS

The same shield protects lines from storms.



Concrete poles and a cable net catch the tree. The line stays up.

HOW IT WORKS · TWO MATERIALS

Concrete goes to the transformer. Cables become the net.

CONCRETE POLES



C1

Intact poles → posts

Set 2 m around the transformer.



C2

Broken poles → crushed

Crushed concrete fills a low base wall.

Concrete never goes to the inventory. It goes **straight from the depot to the site**: posts stand, rubble fills the wall.

CABLES



A1

Into the inventory

Crews throw any offcut into one bin.



A2

Cut to one size

Every cable is cut to 2 m pieces.



A3

Tie into a net

2 m pieces laid 10 cm apart both ways, tied at every crossing: a 2 × 2 m panel.



A4

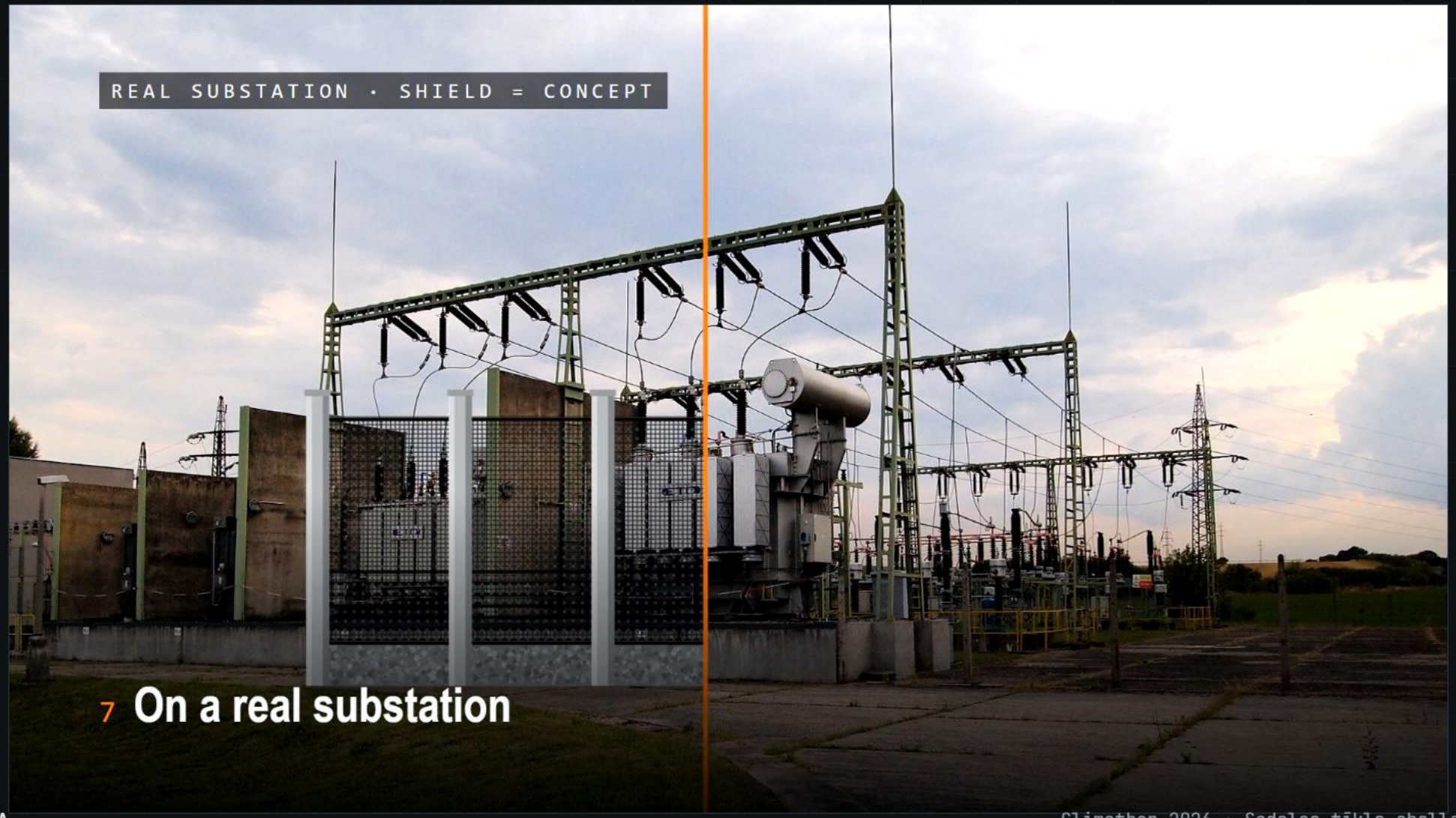
Hang on the posts

Panels clip onto overhead-cable ropes between the posts.

Photos: GeorgHH (CC BY-SA 4.0), S.J. de Waard (CC BY-SA 3.0), Encik Tekateki (CC0), Tiia Monto (CC BY-SA 4.0) with Matrica visualisation.

HOW IT WORKS · 30 SECONDS

From bin to shield.



WHERE

Every critical object we can protect in Latvia.



110 kV substation transformers	33
Grid battery storage	2
Hydro plant transformers	3
Heat and power plant transformers	2
Gas storage and stations	1
Broadcast transmitters	1
Airport navigation aids	2

Points at town level from public sources; exact sites to confirm with AST and Sadales tīkls. Most 110 kV substations have two transformers: ~250–300 in total.

ECONOMICS

One kit. Clear numbers. ESTIMATES

€10–14k
our cost per transformer kit

€18–25k
price per kit

~10%
yearly inspection contract

~€0
material cost: it's their waste

TRY IT · MARKET MODEL

Price per kit: €20k

Transformers protected: 275

Latvia ≈ 250–300 transformers. Baltics ≈ 3× that. Cost assumed €12k per kit.

€5.5M
kit revenue

€2.2M
gross margin

€550k
service per year

↓
disposal cost for the grid company

PRACTICAL

Built with what crews already know.

Safe

Earthed net

Outside the 110 kV clearance zone

Built under the site's work permit

Honest limits

Stops FPV and small drones, debris and falling branches

Does not stop Shahed-class drones or missiles

Version 1 covers the sides

Pilot

2–5 transformers in Latgale

With Sadales tīkls

Measure cost, time, tonnes reused and winter performance

TARGET SUPPLIERS, CUSTOMERS AND PARTNERS

The companies that would want this.

MATERIAL SUPPLIERS AND BUYERS

ST | AST | elektrum group | conexus BALTIC GRID | GASO | LVRTC

Tet | LMT | Tele2 | Bite | LDz | Elektrilevi

elering | Telia | eso | Litgrid | ignitis grupė

DELIVERY PARTNERS

JAUDA | ELTEL | VINCI ENERGIES | Eco Baltia | NKT | prysmian | SAF Tehnika | Origin Robotics

EUROPEAN FUNDS FOR SCALE-UP

Latvian dual-use programme LIAA / ALTUM, EU co-financed

LIFE Circular economy, material reuse

Horizon Europe Cluster 3 Critical infrastructure protection

EIC Accelerator Grant + equity to scale

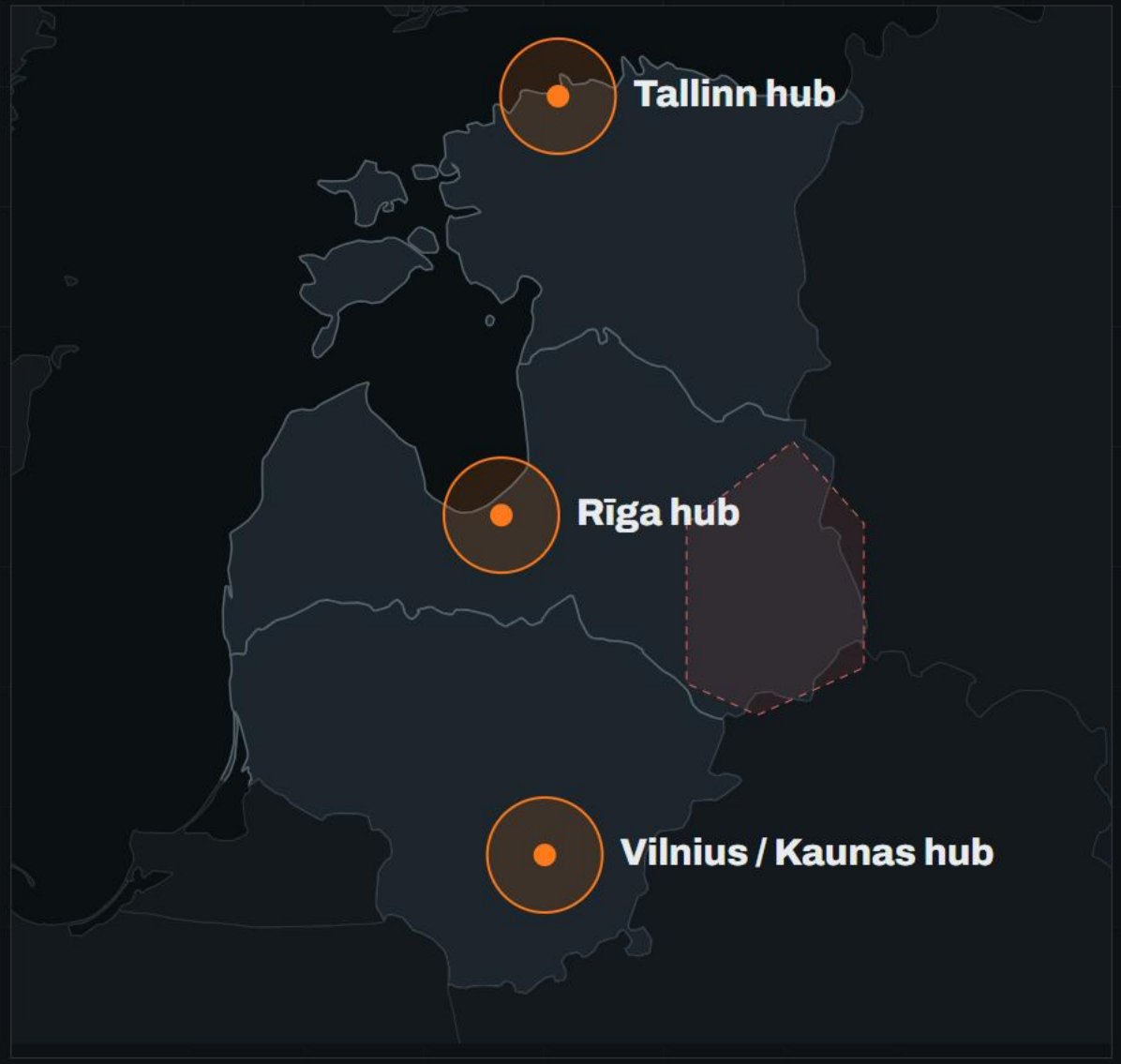
Interreg Baltic Sea Region Three-country rollout

No agreements yet. Logos belong to their owners and are shown to name target companies.

THE FUTURE

A Baltic take-back network.

Three hubs. Every grid, telecom and renewables company as a supplier. Each country's waste protects its own grid.



OUR ASK

A pilot with Sadales tīkls on 2–5 transformers.

APPENDIX

Scored target objects (0–10 each)

Object	Owner	Crit	Fit	Exp	Vol	Safe	Pay	/60
110/20 kV substation transformers	AST / Sadales tīkls	9	8	7	8	5	9	46
20/0.4 kV kiosks feeding hospitals and water	Sadales tīkls	6	10	5	9	8	7	45
Grid battery storage containers	AST	9	9	8	2	7	9	44
Gas regulating and metering stations	Conexus / Gaso	7	9	6	7	8	7	44
Drinking-water pumping stations	Municipal utilities	8	9	5	8	9	5	44
Mobile base station cabinets	LMT / Tet / Tele2 / Bite	7	8	6	10	8	5	44
Broadcast transmitter sites	LVRTC	8	7	6	4	7	7	39
Railway traction substations	LDz	7	8	6	5	6	6	38
Small district-heating boiler houses	Municipal	7	6	5	7	8	5	38
Data-centre cooling and generator yards	Tet and others	7	8	4	4	8	7	38
Synchronous condensers	AST	9	5	6	2	6	9	37
Airport navigation aids	LGS	8	7	4	3	8	7	37

Assumptions behind the economics

- Kit for one 110/20 kV transformer: ~14 concrete posts, ~180 m^2 of cable net, ~120 m AMKA rope, ~10 m^3 crushed-concrete wall.
- Cost €10–14k: baskets and earthing €2–3k, cable cutting and tying €2–3k, installation 3 people × 4 days €4–5k, engineering and transport €2–3k.
- Price €18–25k and a yearly service at ~10% are hypotheses to validate with Sadales tīkls.
- ~250–300 transformers in Latvia (most 110 kV substations have two). To confirm with AST and Sadales tīkls.

Sources

- Sadales tīkls challenge brief and stock list
  `Climathon 2026 RTU materials (this repo)`
- Ukraine: three levels of energy protection
  `https://war.ukraine.ua/war-news/ukraine-protection-russias-attacks-energy-infrastructure/`
- EIB €86M to Ukrenergo for anti-drone protection
  `https://en.interfax.com.ua/news/general/1036088.html`
- Ukraine netted 822 km of roads in 2025
  `https://www.yahoo.com/news/world/articles/ukraine-lined-roads-enough-anti-062209532.html`
- Lithuania's Gižai substation, €48M+ hardening
  `https://eutoday.net/lithuania-drone-proof-harmony-link-eu-grid-security/`
- AST batteries in Rēzekne and Tume
  `https://www.ess-news.com/2025/10/30/latvia-adds-big-batteries-to-complete-grid-sync-with-europe-two-major-estonian-projects-secure-finance`
- Anti-drone net mesh sizes and twine ratings
  `https://ahcebrian.com/en/blog/passive-anti-drone-nets-explained/`
- EU CER Directive, Latvian transposition
  `https://www.iem.gov.lv/en/article/government-strengthens-critical-infrastructure-resilience-and-national-security`
- Baltic defence and dual-use investors
  `https://www.vestbee.com/insights/articles/europes-defence-and-dual-use-investors`
- Full research with all sources
  `research/grid-shield.html`

Image and data credits

- **Cable scrap pile:** GeorgHH, CC BY-SA 4.0, Wikimedia Commons: Elektrokabelschrott 0109.jpg
- **Substation before/after:** Tiia Monto, CC BY-SA 4.0, Wikimedia Commons: Milovice-transformer station.jpg. The after image is an AI edit with Qwen-Image-Edit, shared under CC BY-SA 4.0.
- **110/20 kV transformer:** Kreuzschnabel, CC BY-SA 3.0, Wikimedia Commons: Transformer 110 kV 20 kV.jpg
- **Storm damage, Arlington:** Arlington County, CC BY-SA 2.0, Wikimedia Commons: Storm Downed power lines and trees from Storm Sandy (8138919297).jpg
- **Gabion wall:** Encik Tekateki, CC0, Wikimedia Commons: Gabion Wall.jpg
- **Hydraulic cable cutter:** S.J. de Waard, CC BY-SA 3.0, Wikimedia Commons: Cable cutting device.JPG
- **Country outlines:** mledoze/countries (GitHub), ODbL, from Natural Earth
- **Company logos:** Wikimedia Commons and the companies' own websites. Logos belong to their owners.
- **Explainer video:** Matrica, made with Remotion (github.com/remotion-dev/remotion)